Fake News Detection Using Machine Learning: A Comparative Study of Multiple Models

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K E Y W O R D S

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Support Vector Machine
Logistic Regression
Data Processing
Generalization

A B S T R A C T

Fake news has become a major issue in today's digital world, spreading misinformation rapidly across various platforms. This project focuses on detecting fake news using machine learning techniques. Multiple models including Logistic Regression, Naive Bayes, and Support Vector Machine (SVM) were implemented and evaluated on different datasets. Initial experiments on a basic dataset resulted in low accuracy (around 60%), highlighting issues such as data imbalance and poor feature representation. Further improvements using optimized models increased accuracy to approximately 65%. A large structured dataset achieved very high accuracy (around 99%), but showed limited generalization to real-world inputs. To address this, a custom dataset was designed to simulate real-world scenarios, resulting in a balanced performance with an accuracy of approximately 83% and better handling of random inputs. The study highlights the importance of dataset quality and demonstrates the trade-off between accuracy and generalization in machine learning models for fake news detection.

1 Introduction

In the digital age, the rapid growth of online platforms and social media has made the spread of information faster and more accessible than ever before. However, this has also led to a significant rise in the circulation of fake news, which can mislead people, create confusion, and even influence public opinion and decision-making. Detecting fake news has therefore become an important challenge in the field of data science and machine learning.

Traditional methods of verifying news are often time-consuming and require human intervention, making them less effective in handling the large volume of information generated daily. As a result, automated approaches using machine learning have gained popularity for identifying and classifying fake news efficiently.

This project aims to explore the use of different machine learning models for fake news detection. By analyzing and comparing multiple datasets and models, the study focuses on understanding how data quality and structure impact model performance. The project also highlights the challenges of generalization, where a model performs well on structured datasets but struggles with real-world inputs. Through this approach, the research provides insights into building more reliable and adaptable fake news detection systems.

2 Objectives

The main objectives of this project are as follows:

* To develop a machine learning model for detecting fake and real news.

* To analyze the impact of different datasets on model performance.

* To compare the effectiveness of various algorithms such as Logistic Regression, Naive Bayes, and Support Vector Machine (SVM).

* To identify challenges such as data imbalance, overfitting, and lack of generalization.

* To design a custom dataset that better represents real-world scenarios.

* To improve the model's ability to handle short and diverse input statements.

* To study the trade-off between accuracy and generalization in fake news detection systems.

3 Dataset Discription

This project utilizes three different datasets to evaluate the performance of machine learning models for fake news detection. Each dataset serves a specific purpose in understanding model behavior and improving overall performance.

3.1 **Dataset 1 (Initial Dataset):**

The first dataset used in this study was relatively small and imbalanced, with a higher proportion of fake news samples compared to real news. This imbalance led to biased model predictions, where most inputs were classified as fake. The

limited size and lack of diversity in the dataset resulted in poor performance, with an accuracy of approximately 60%.

3.2 **Dataset 2 (Structured Dataset):**

The second dataset was obtained from a publicly available source and contained a large number of well-structured news articles. It was more balanced and included detailed textual information, which helped the model achieve very high accuracy (around 99%). However, the dataset had a consistent format and writing style, which caused the model to rely on specific patterns. As a result, the model showed poor generalization when tested on short or informal real-world inputs.

3.3 **Dataset 3 (Custom Dataset):**

To overcome the limitations of the previous datasets, a custom dataset was designed. This dataset included short and medium-length sentences representing both real and fake news scenarios. It was carefully balanced and structured to simulate real-world inputs. Although the accuracy achieved on this dataset was slightly lower (around 83%), the model demonstrated significantly better generalization and was able to correctly classify random and diverse statements.

Overall, the use of multiple datasets allowed a comprehensive analysis of model performance, highlighting the importance of dataset quality, balance, and diversity in building reliable fake news detection systems.

4 Methodology

The methodology of this project involves a systematic approach to building and evaluating machine learning models for fake news detection. The complete process is divided into the following steps:

4.1 Data Collection:

Multiple datasets were used in this project, including an initial basic dataset, a large structured dataset, and a custom-designed dataset. This helped in analyzing model performance under different data conditions.

4.2 Data Preprocessing:

The text data was cleaned and prepared for analysis. This included converting all text to lowercase to maintain consistency and reduce variation caused by different text formats.

4.3 Data Splitting:

The dataset was divided into training and testing sets using an 80:20 ratio. Stratified sampling was applied to ensure that both classes (real and fake) were equally represented in the training and testing data.

4.4 Feature Extraction:

Text data was converted into numerical form using the TF-IDF (Term Frequency–Inverse Document Frequency) technique. This helped in identifying the importance of words in the dataset and enabled machine learning models to process textual information effectively.

4.5 Model Training:

Different machine learning models such as Logistic Regression, Naive Bayes, and Support Vector Machine (SVM) were trained on the processed data. Each model was evaluated to understand its performance and limitations.

4.6 Model Evaluation:

The performance of each model was measured using accuracy as the primary metric. Additionally, predictions were tested on custom inputs to evaluate how well the model generalizes to real-world scenarios.

4.7 Improvement and Optimization:

Several improvements were applied, including handling data imbalance, tuning model parameters, and refining the dataset. A custom dataset was also created to enhance the model's ability to handle diverse and short inputs.

This step-by-step methodology ensured a comprehensive evaluation of different models and highlighted the importance of both data quality and model selection in fake news detection.

5 Models Used

In this project, three different machine learning models were implemented and evaluated for fake news detection. Each model was selected based on its suitability for text classification tasks and its ability to handle different types of datasets.

5.1 Logistic Regression:

Logistic Regression is a simple and widely used classification algorithm that works well for binary classification problems. It predicts the probability of a given input belonging to a particular class. In this project, it was used as a baseline model to understand the initial performance on the dataset. However, due to data imbalance and limited feature representation, it achieved relatively low accuracy (around 60%).

5.2 Naive Bayes:

Naive Bayes is a probabilistic classifier based on Bayes' Theorem, which assumes that features are independent of each other. It is particularly effective for text classification tasks because it handles word frequencies efficiently. In this project, Naive Bayes performed better than Logistic Regression and achieved an accuracy of approximately 65% after improvements in preprocessing and feature extraction.

5.3 Support Vector Machine (SVM):

Support Vector Machine is a powerful supervised learning algorithm used for classification tasks. It works by finding an optimal boundary (hyperplane) that separates different classes. In this project, SVM provided the highest accuracy (around 99%) when trained on a structured dataset. However, it showed limitations in generalizing to real-world inputs due to its reliance on specific data patterns.

By comparing these models, the project highlights how different algorithms perform under varying data conditions and emphasizes the importance of selecting the right model based on the dataset and problem requirements.

6 Results And Analysis

The results of this project highlight the impact of dataset quality, model selection, and preprocessing techniques on the performance of fake news detection systems. Multiple models were evaluated on different datasets to analyze their behavior and effectiveness.

Table 1- Performance Comparison of Machine Learning Models

Models	Dataset	Accuracy	Observation
Logistic Regression	Initial	60%	Poor , Biased
Naïve Bayes	Improved	65%	Slight better
SVM	Structured	99%	Overfitting
Naïve Bayes	Custom	83%	Best generalization

The initial model using Logistic Regression showed poor results, mainly due to the imbalance and limited size of the dataset. Most of the predictions were biased towards fake news, indicating that the model was not able to learn meaningful patterns.

After applying improvements and using Naive Bayes, the performance increased slightly. This demonstrated that preprocessing and feature extraction techniques such as TF-IDF can enhance model accuracy, but the dataset limitations still affected the results.

A significant improvement was observed when using a large structured dataset with the SVM model, achieving an accuracy of approximately 99%. However, further testing revealed that the model relied heavily on specific patterns and formatting present in the dataset. As a result, it failed to generalize effectively to short and informal inputs.

To address this issue, a custom dataset was created with diverse and realistic sentences. Although the accuracy decreased to around 83%, the model demonstrated much better generalization and was able to correctly classify random real-world inputs. This indicates that a slightly lower accuracy can still represent a more practical and reliable model.

Overall, the analysis shows that while high accuracy is desirable, it does not always guarantee real-world performance. The balance between accuracy and generalization is crucial in building effective fake news detection systems.

7 Challenges Faced

During the development of this project, several challenges were encountered while building and evaluating the fake news detection models. These challenges helped in gaining a deeper understanding of machine learning concepts and model behavior.

7.1 Data Imbalance:

The initial dataset was imbalanced, with a higher number of fake news samples compared to real news. This caused the model to become biased and predict most inputs as fake, resulting in poor performance.

7.2 Poor Model Performance:

In the early stages, the models achieved low accuracy (around 60%) and failed to correctly classify real news. This highlighted the limitations of using small and low-quality datasets.

7.3 Model Bias:

Even after improvements, the models tended to favor one class (fake news), indicating that the learning was not well-balanced.

7.4 Overfitting on Structured Dataset:

When using a large structured dataset, the model achieved very high accuracy (around 99%) but relied heavily on specific patterns and formats present in the data. This led to overfitting and poor generalization to real-world inputs.

7.5 Generalization Issue:

The model struggled to correctly classify short and random sentences, as it was trained mostly on long and formal news articles.

7.6 Small Custom Dataset:

While creating the custom dataset improved generalization, the limited number of samples initially resulted in low accuracy (around 50%), which required further expansion of the dataset.

7.7 Debugging and Implementation Issues:

Several technical issues were faced during implementation, such as incorrect data splitting, vectorization errors, and inconsistent predictions. These required careful debugging and step-by-step validation of the workflow.

Overcoming these challenges played a crucial role in improving the final model and provided valuable insights into the importance of data quality, preprocessing, and proper model evaluation

8 Conclusion

In conclusion, this project successfully explored the use of machine learning techniques for fake news detection by implementing and comparing multiple models across different datasets. The study demonstrated that while models such as Support Vector Machine can achieve very high accuracy on structured datasets, they may fail to generalize effectively to real-world inputs.

The results highlighted the importance of dataset quality, balance, and diversity in building reliable models. Initial experiments with small and imbalanced data resulted in poor performance, while structured datasets improved accuracy but introduced overfitting issues. The development of a custom dataset helped address these limitations by improving the model's ability to handle short and diverse inputs, even though the overall accuracy was slightly lower.

Overall, the project emphasizes that achieving high accuracy alone is not sufficient. A well-performing model should also be capable of generalizing to real-world scenarios. This work provides valuable insights into the trade-offs between accuracy and generalization and demonstrates the importance of thoughtful dataset design in machine learning applications.

9 Future Scope

The future scope of this project includes several improvements to enhance the performance and applicability of fake news detection systems. One of the key areas of improvement is the use of larger and more diverse datasets, which can help the model learn better patterns and improve generalization.

Advanced machine learning and deep learning techniques, such as Recurrent Neural Networks (RNN), Long Short-Term Memory (LSTM), and transformer-based models, can be explored to achieve more accurate and context-aware predictions. These models are capable of understanding complex language patterns and can significantly improve performance on real-world data.

Another potential improvement is the integration of real-time data from social media platforms and news websites, which would allow the system to detect fake news dynamically. Additionally, incorporating features such as source credibility, user behavior, and metadata can further enhance the reliability of the model.

The custom dataset can also be expanded by including more diverse examples, different writing styles, and multilingual data to make the system more robust and adaptable. Finally, deploying the model as a web or mobile application can make it accessible to users, enabling practical use in everyday scenarios.

Overall, there is significant scope for enhancing both the accuracy and real-world applicability of fake news detection systems through advanced techniques and improved data.

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11 References

The following resources were referred to during the development of this project:

• Kaggle Dataset: Fake and Real News Dataset by Clément Bisaillon

• Scikit-learn Documentation: Machine Learning Library for Python

• Google Colab: Cloud-based platform for executing Python code

• Various online tutorials and educational videos on machine learning and fake news detection

• Research articles and blogs related to Natural Language Processing (NLP) and text classification techniques

These resources provided guidance for data handling, model implementation, and understanding of machine learning concepts used in this project.